A MORPHISM OF ALGEBRAIC SPACES IS INTEGRAL EXACTLY WHEN IT IS AFFINE AND UNIVERSALLY CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has two enumerated equivalent conditions; the proof verifies that both may be tested after the same base change and then cites the scheme case.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Locality of integrality, affineness and universal closedness. Lemmas lemma-integral-local, lemma-affine-local and lemma-universally-closed-local, used at proof lines 9458-9462. Reduce both sides of the equivalence to the affine case.
- The scheme case: integral equals affine plus universally closed. Morphisms, Lemma morphisms-lemma-integral-universally-closed, used at proof lines 9463-9464. Is the input that closes the argument.
- Representability of the morphisms involved. used at proof lines 9458-9459. Makes the base change reduction available.

Source reference. Original lemma 03WG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5111–5111. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03ZQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5229–5229. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03IT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4615–4615. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2487–2487. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent*

(1) *f is integral, and*
(2) *f is affine and universally closed.*

Proof. In both cases the morphism is representable, and you can check the condition after a base change by an affine scheme mapping into Y, see Lemmas 03ZQ, 03WG, and 03IT. Hence the result follows from Morphisms, Lemma 01WM. □